

The baseline statistical predictors used by **SARTA** (Stand-alone AIRS Radiative Transfer Algorithm) are mathematical combinations of four primary physical state variables: **layer temperature (T)**, **target gas absorber amount (u)**, **pressure (P)**, and **path angle secant (θ)**. [1, 2]

Instead of passing raw variables directly, SARTA uses an expansion of these inputs into a 10-to-15-term polynomial suite for each atmospheric layer. This is done to linearize the highly non-linear physics of molecular line absorption. [2, 3]

The exact predictors are divided into category groups based on the type of gas being calculated:

1. Fixed Gas Predictors (For well-mixed gases like CO₂, O₂, N₂O)

Because well-mixed gases have relatively constant ratios in a layer, their predictors isolate the effects of temperature variations away from a reference standard profile (T_{ref}):

- δT : The absolute difference between the actual layer temperature and the reference profile temperature ($T - T_{ref}$).
- $(\delta T)^2$: The squared temperature variance, mapping non-linear thermal Doppler broadening.
- $\sec(\theta)$: The secant of the local viewing zenith angle, scaling the geometric optical path length through the layer.
- $\delta T \cdot \sec(\theta)$: An interaction term capturing path-dependent temperature changes. [2]

2. Water Vapor Predictors (H₂O)

Water vapor is highly variable and dominates infrared absorption, requiring a more intense set of predictors that blend temperature offsets with water mass ratios (Q):

- Q / Q_{ref} : The ratio of the actual water vapor column density to the reference standard profile.
- $\delta T \cdot (Q / Q_{ref})$: Tracks how water vapor absorption behavior shifts with temperature.
- $(Q / Q_{ref}) \cdot \sec(\theta)$: Combines local water mass abundance with the slanted viewing path.
- $\sqrt{Q / Q_{ref}}$: Evaluates the square root of the column ratio to linearize strong, saturated water lines.

3. Ozone Predictors (O₃)

Similar to water vapor, ozone varies dramatically by altitude layer (especially in the stratosphere) and uses concentrated ratio expansions:

- V / V_{ref} : The ratio of the actual layer ozone amount to the reference profile.
- $(V / V_{ref}) \cdot \sec(\theta)$: Scales ozone abundance along the slant path.

- $\Delta T \cdot (V / V_{\text{ref}})$: Evaluates the temperature dependence of specific ozone vibrational-rotational bands.

4. Cloud Scattering Predictors (When active)

If SARTA is run in its cloudy configurations, the regression framework adds macroscopic cloud physical metrics to calculate scattering scaling factors: [1, 3]

- Cloud Top Pressure (P_{top})
- Cloud Optical Depth (τ)
- Effective Particle Radius (r_e) [1, 4]

How They Synthesize an Output

SARTA evaluates these exact polynomial terms for every layer. It then multiplies the results by a pre-calculated matrix of **regression coefficients (A)** derived from line-by-line forward models (like LBLRTM) to output the final channel optical depth (τ_{channel}): [1, 2]

$$\tau_{\text{channel}} = \sum_{i=1}^N A_i \cdot \text{Predictor}_i$$

Would you like to see the **specific mathematical polynomial string** for a fixed gas layer, or explore how SARTA generates its **reference atmospheric profiles**?

[1] <https://agupubs.onlinelibrary.wiley.com>

[2] <https://www.ecmwf.int>

[3] <https://www.ecmwf.int>

[4] <https://www2.mmm.ucar.edu>